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(54) COLOR DISPLAY WITH COLOR FILTER LAYER COMPRISING TWO-DIMENSIONAL PHOTONIC CRYSTALS FORMED IN A DIELECTRIC LAYER

(71) Applicant: **Taiwan Semiconductor Manufacturing Company, LTD.**, Hsinchu (TW)

(72) Inventors: **Hong-Shyang Wu**, Taipei (TW); **Kuo-Ming Wu**, Hsinchu (TW)

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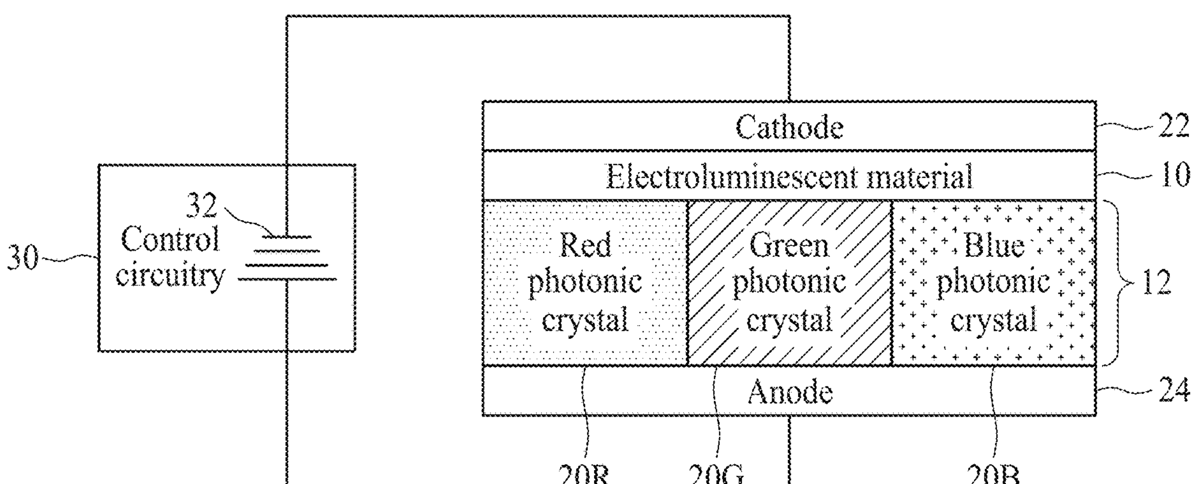
(52) **U.S. Cl.**

CPC ***F21V 9/40*** (2018.02); ***G02F 1/134309*** (2013.01); ***G02F 1/13439*** (2013.01); ***G02F 1/1347*** (2013.01)

(57) **ABSTRACT**

In a color display, a color filter layer includes a dielectric layer with an array of photonic crystals, an electroluminescent material disposed on the color filter layer, and electrodes arranged to electrically energize the electroluminescent material to output white light. Each photonic crystal includes a two-dimensional (2D) array of features. The 2D array of features includes a central cavity within which the features of the 2D array of features are omitted. Each photonic crystal is tuned to a resonant wavelength by a periodicity of the two-dimensional array of features. The array of photonic crystals may include, for example, red, green, and blue photonic crystals arranged to form an array of pixels spanning a display area of the color display, in which each pixel includes at least one red photonic crystal, at least one green photonic crystal, and at least one blue photonic crystal.



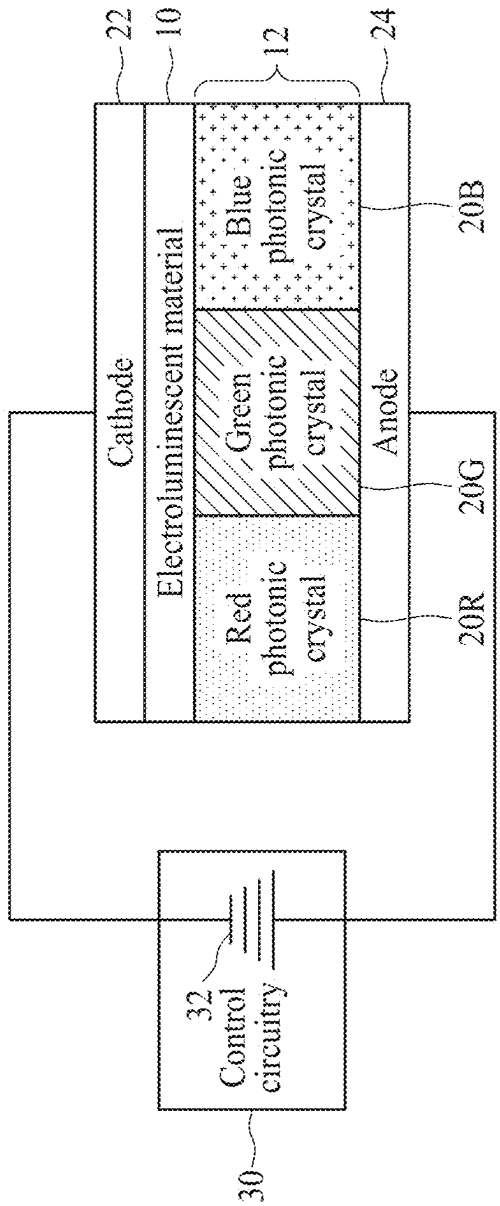

22
10
12
24
Cathode
Electroluminescent material
Blue photonic crystal
Green photonic crystal
Red photonic crystal
Anode
20B
20G
20R
32
Control circuitry
30

FIG. 1

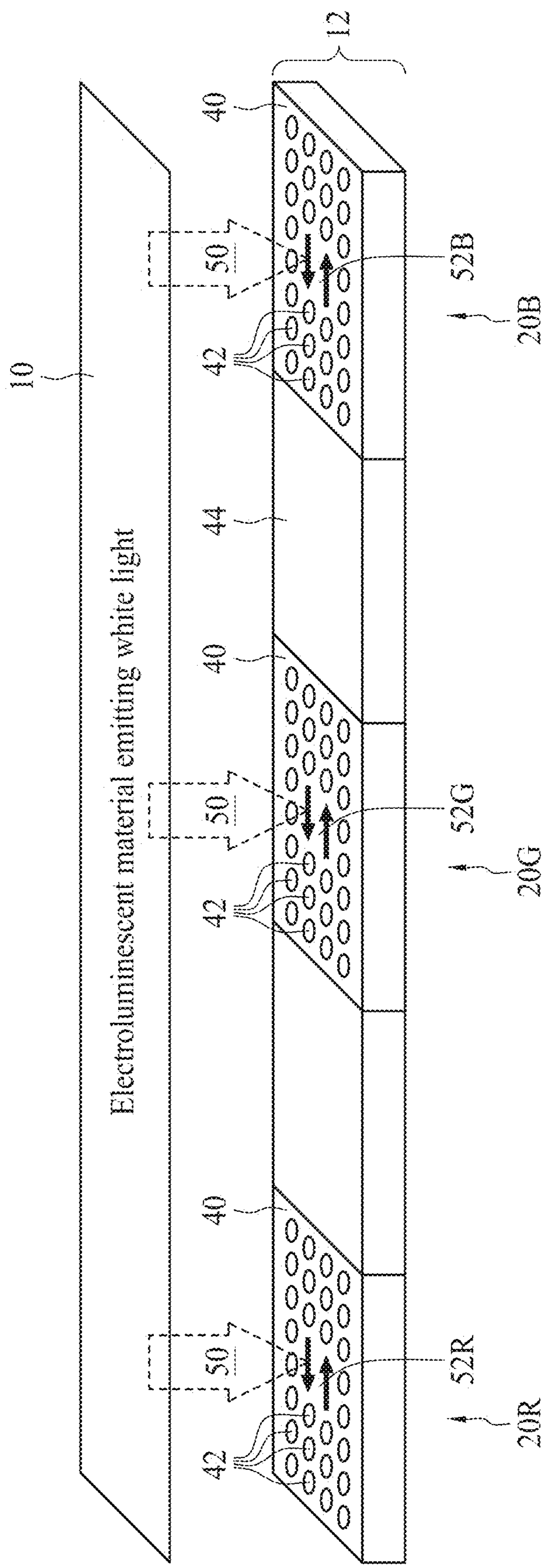

12
10
Electroluminescent material emitting white light
40
50
52B
42
20B
44
40
50
52G
42
20G
40
50
52R
42
20R

FIG. 2

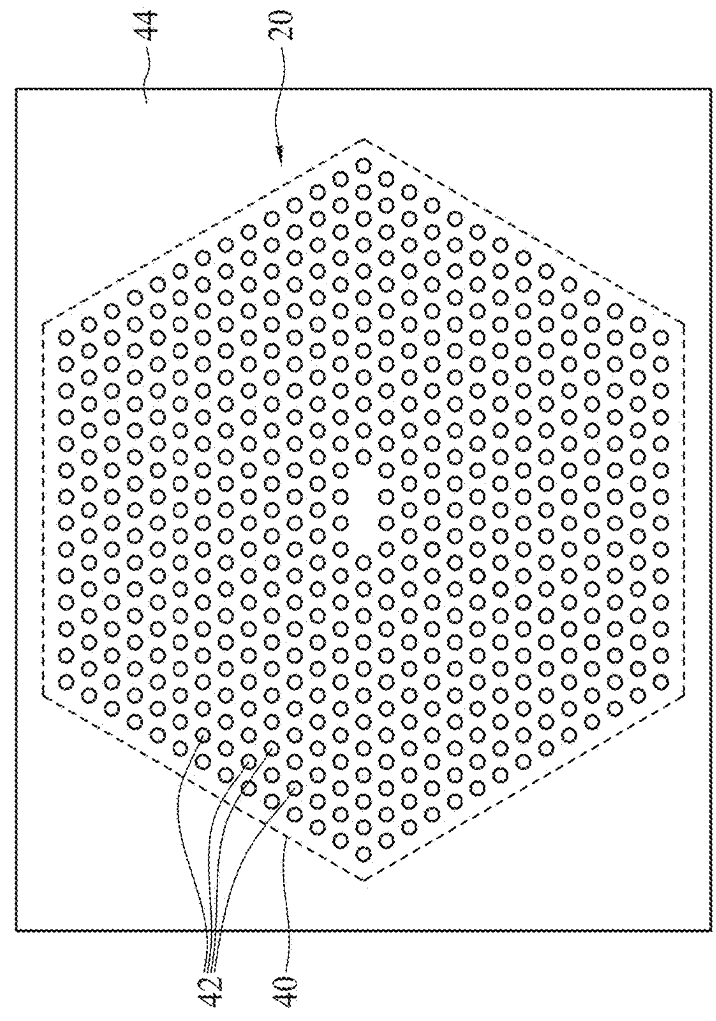


FIG. 3

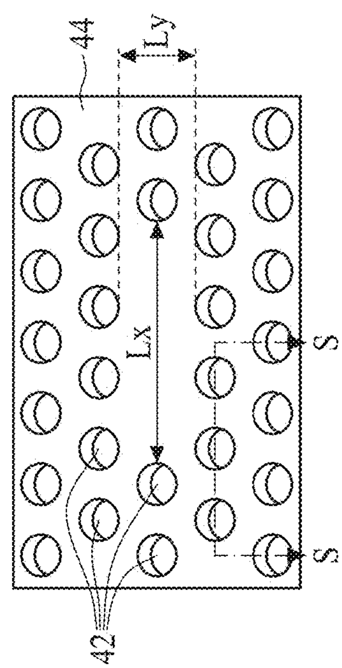


FIG. 4

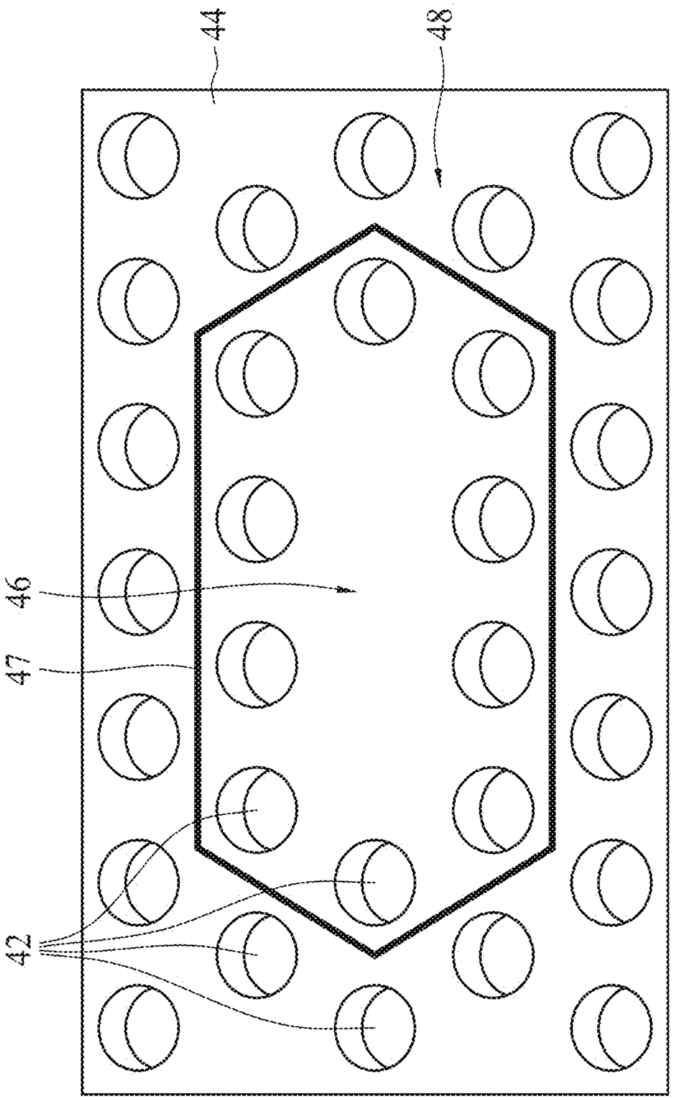


FIG. 5

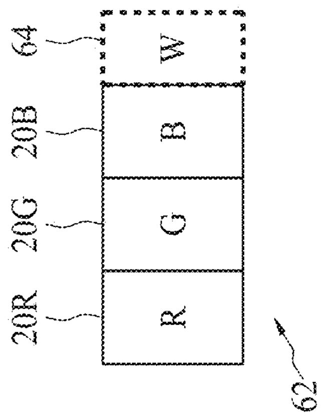


FIG. 7

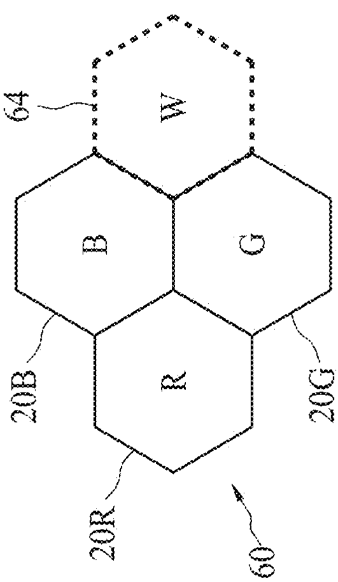


FIG. 6

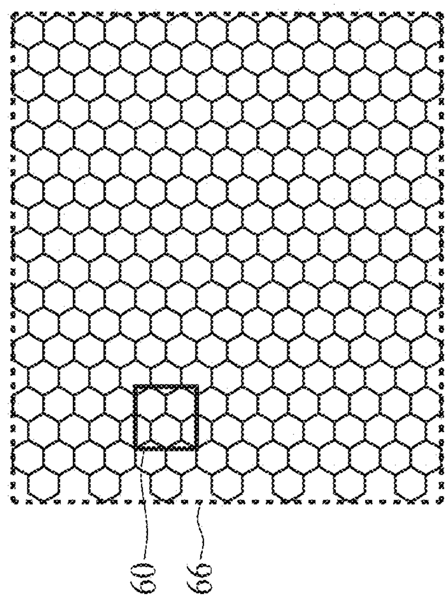


FIG. 8

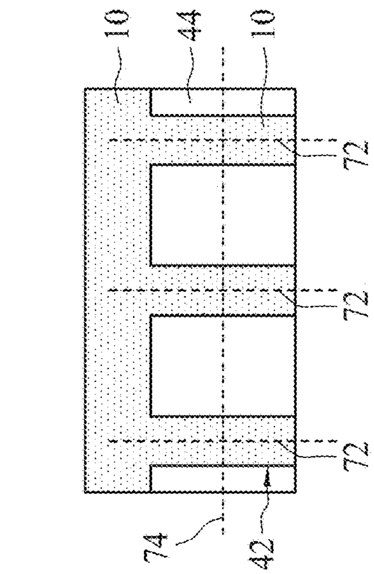


FIG. 11

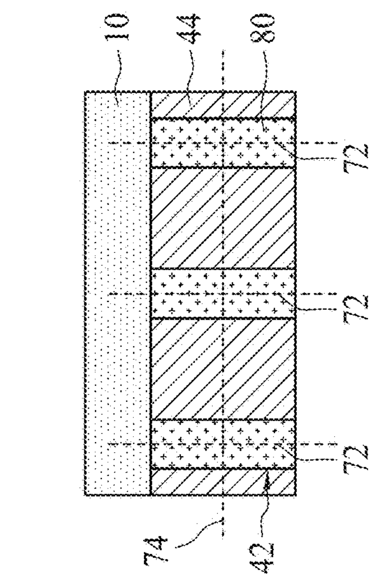


FIG. 10

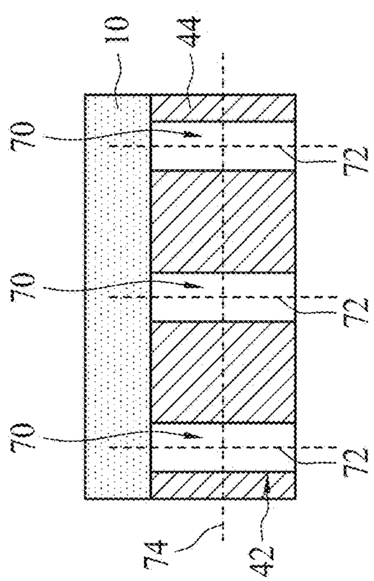


FIG. 9

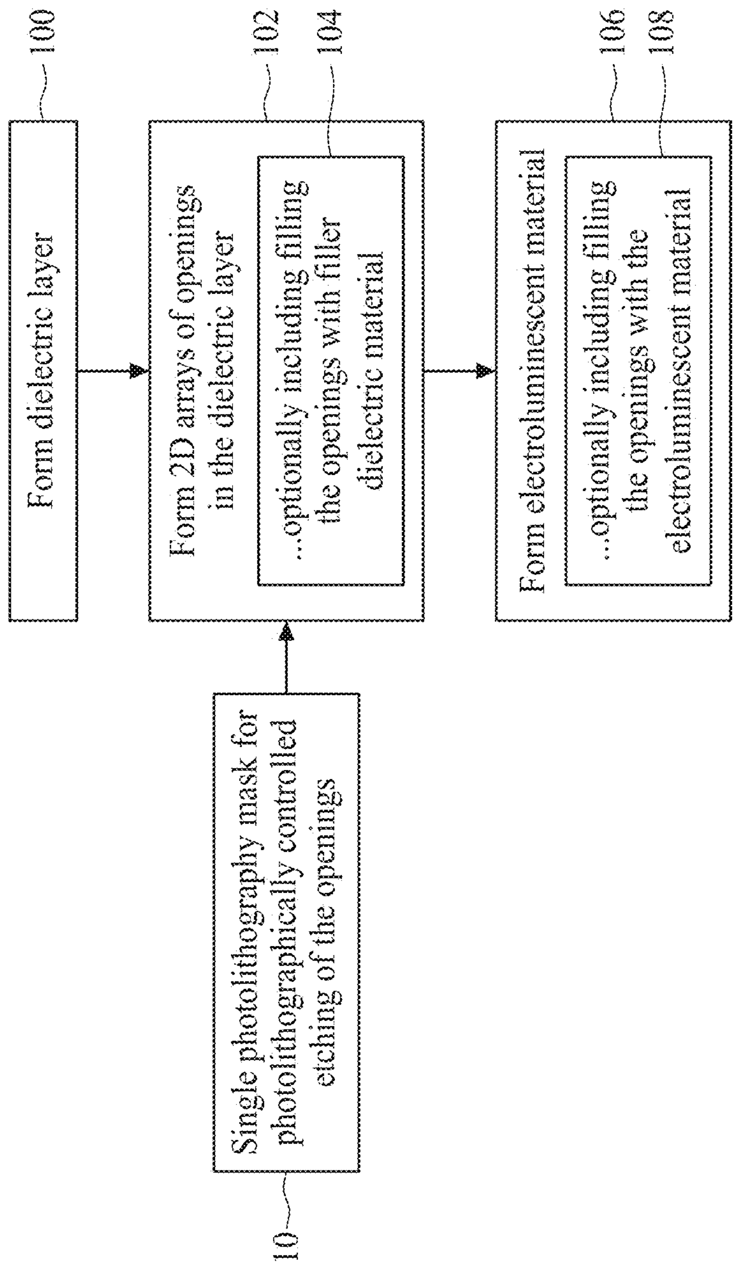

100
Form dielectric layer
102
Form 2D arrays of openings in the dielectric layer
104
...optionally including filling the openings with filler dielectric material
106
Form electroluminescent material
108
...optionally including filling the openings with the electroluminescent material
110
Single photolithography mask for photolithographically controlled etching of the openings

FIG. 12

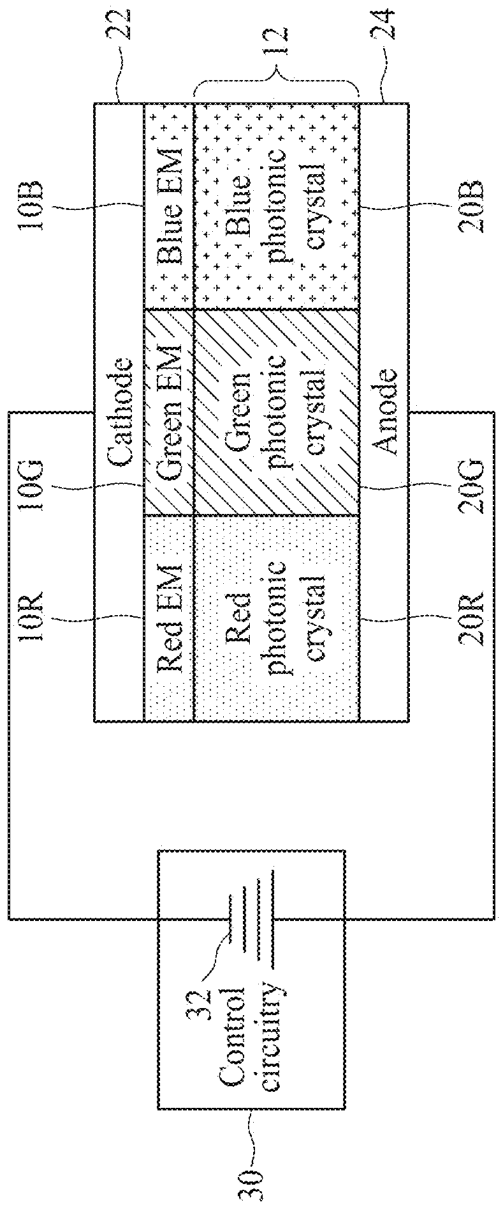

22
12
24
10B
Blue EM
Blue photonic crystal
20B
Cathode
Green EM
Green photonic crystal
Anode
10G
20G
10R
Red EM
Red photonic crystal
20R
32
Control circuitry
30

FIG. 13

COLOR DISPLAY WITH COLOR FILTER LAYER COMPRISING TWO-DIMENSIONAL PHOTONIC CRYSTALS FORMED IN A DIELECTRIC LAYER

[0001] This application is a continuation of U.S. application Ser. No. 18/231,897 filed Aug. 9, 2023, which is a continuation of U.S. application Ser. No. 17/749,295 filed May 20, 2022, which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] The following relates to the color display arts, organic light emitting diode (OLED) display arts, quantum dot (QD) display device arts, active matrix OLED (AMOLED) display arts, and related arts.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. **1** diagrammatically illustrates a high level side-sectional view of a color display employing an electroluminescent material emitting white light and a color filter layer made up of photonic crystals comprising two-dimensional (2D) arrays of features formed in a dielectric layer.

[0005] FIG. **2** diagrammatically illustrates operation of a color filter layer made up of photonic crystals comprising 2D arrays of features formed in a dielectric layer.

[0006] FIG. **3** diagrammatically illustrates a top view of a photonic crystal comprising a 2D array of features formed in a dielectric layer.

[0007] FIG. **4** diagrammatically illustrates an enlarged perspective view of a central portion of the photonic crystal of FIG. **3**.

[0008] FIG. **5** diagrammatically illustrates the same enlarged perspective view of the central portion of the photonic crystal of FIG. **3** as is shown in FIG. **4**, but with an inner cavity and outer zone delineated.

[0009] FIGS. **6** and **7** diagrammatically illustrate two layout embodiments for red, green, and blue photonic crystals forming a pixel of a color display.

[0010] FIG. **8** diagrammatically illustrates an array of pixels as shown in FIG. **6** spanning a display area of a color display.

[0011] FIGS. **9**, **10**, and **11** diagrammatically illustrate Section S-S indicated in FIG. **4**, according to three respective embodiments.

[0012] FIG. **12** shows a flow chart of a suitable manufacturing method for forming a multicolor light emission structure of a color display employing an electroluminescent material emitting white light and including a color filter layer made up of photonic crystals comprising 2D arrays of features formed in a dielectric layer.

[0013] FIG. **13** diagrammatically illustrates a high level side-sectional view of a color display employing red-emitting, green-emitting, and blue-emitting electroluminescent materials and a color filter layer made up of photonic crystals comprising 2D arrays of features formed in a dielectric layer.

DETAILED DESCRIPTION

[0014] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0015] Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0016] The following relates to color displays of types employing an electroluminescent material optically coupled with an array of color filters of at least three different colors. For a full-color display, the three different filter colors are suitably primary colors, typically red, green, and blue for a light-emissive color display. Embodiments disclosed herein employ a color filter layer made up of photonic crystals comprising two-dimensional (2D) arrays of features formed in a dielectric layer. The disclosed color displays employing color filters in accordance with this approach provide numerous advantages such as low cost manufacturing with a small number of workflow steps or operations, improved planarity of the multicolor light emission structure of a color display, facility for high precision and run-to-run reproducibility in defining the emission wavelengths, and facilitation of scalability to smaller pixel size (and hence higher resolution display), in some embodiments entailing only employing a different photomask to achieve a multicolor light emission structure with a different (e.g. higher) pixel resolution.

[0017] With reference to FIG. **1**, a high level side-sectional view is diagrammatically illustrated of a color display employing a multicolor emission structure that includes an electroluminescent material **10** designed to emit white light, and a color filter layer **12** made up of photonic crystals **20**R, **20**G, **20**B comprising 2D arrays of features formed in a dielectric layer. The photonic crystals are tuned to red, green, and blue primary colors, i.e. the photonic crystals **20**R are tuned to red, the photonic crystals **20**G are tuned to green, and the photonic crystals **20**B are tuned to blue. Furthermore, electrodes **22**, **24** are arranged to electrically energize the electroluminescent material **10** to output light. In the example of FIG. **1**, the electroluminescent material **10** outputs white light including at least all three primary (e.g. red, green, and blue) color components. (More typically, the output white light has a spectrum extending over most or all of the visible range, i.e. 400-700 nm wavelength).

[0018] In some embodiments, the electroluminescent material is an organic electroluminescent diode (OLED) material. In some embodiments, the electroluminescent material is a colloidal quantum dot (QD) electroluminescent material. Other types of electroluminescent materials are also contemplated as thew electroluminescent material **10**. In the illustrative examples, the electroluminescent material **10** is an OLED material, and the electrodes **22**, **24** include a cathode **22** and an anode **24**. It is to be understood that FIG. **1** is a high-level diagrammatic representation, and the multicolor emission structure **10**, **12** and associated electrodes **22**, **24** may include numerous additional layers or components not shown. For example, in the case of an OLED color display in which the electroluminescent material **10** is an OLED material, such additional layers or components may for example include an electron injection or transport layer (EIL or ETL) disposed between the cathode **22** and the emission structure **10**, **12**, and/or a hole transport or injection layer (HTL or HIL) disposed between the anode **24** and the emission structure **10**, **12**. Moreover, the ordering of the stack of layers shown in FIG. **1** could be varied—for example, it is contemplated for one of the electrodes **22**, **24** to be interposed between the electroluminescent material **10** and the color filter layer **12**. The overall structure may be disposed on (and optionally formed on) a substrate made of glass, sapphire, or another material, and various additional layers may be provided as interface modification layers or the like.

[0019] These additional components may be configured to structure the color display as a bottom-emitting display in which the display is observed from the substrate-side (in which case the light passes through the substrate which should be transparent over the spectrum of interest, e.g. over the visible spectrum), or as a top-emitting display in which the display is observed from the side opposite the substrate. For the OLED display example, the bottom-emitting configuration is sometimes referred to as a bottom-emitting OLED display (BEOLED) and the top-emitting configuration is sometimes referred to as a top-emitting OLED display (TEOLED). In general, the configuration for bottom- or top-emission includes making the electrode on the emission side optically transparent over the visible spectrum (for example, made of indium tin oxide, i.e. ITO or a sufficiently thin metal to be light-transmissive) and making the electrode on the side opposite the emission side optically reflective over the visible spectrum (for example, made of gold or another metal of sufficient thickness to be optically opaque).

[0020] Furthermore, the color display typically includes control circuitry **30** configured to selectively apply an electrical bias **32** (e.g., voltage or current) to selected pixels of the display, or to sub-pixels of particular color(s) of the selected pixels, to form a desired color image. The control circuitry **30** is diagrammatically shown in FIG. **1** separately from the optical stack, but in some implementations is constructed as an electronics layer incorporated into the optical stack (i.e., the stack including the electroluminescent material **10** and the color filter layer **12**, along with the electrodes **22**, **24**). By way of some nonlimiting illustrative examples, the control circuitry **30** may implement an active matrix addressing scheme. For example, such active matrix control circuitry may comprise a thin-film transistor (TFT) array in which the TFTs are, for example, polycrystalline silicon (poly-Si) or amorphous silicon (a-Si) TFTs, together with at least one of the electrodes **22**, **24** being segmented to form addressable pixels each of which is independently electrically biased (or not biased, depending on the content being displayed) by a corresponding TFT of the array of TFTs. In the case of such active matrix addressing and using an OLED material as the electroluminescent material **10**, the color display may be referred to as an active-matrix OLED display (AMOLED display). In other embodiments, the control circuitry **30** may implement a passive matrix addressing scheme. The control circuitry **30** typically forms a backplane (e.g., a TFT backplane) that is disposed on the side of the light emission structure (e.g. comprising the electroluminescent material **10** and layer of photonic crystals **20**R, **20**G, **20**B) opposite from the light-emission side. In some embodiments, the TFT backplane is fabricated on a glass, sapphire or other substrate, and the optical stack **10**, **12** and electrodes **22**, **24** are then fabricated on the TFT backplane, although other fabrication sequences are contemplated such as separately fabricating the TFT backplane and the optical stack and subsequently bonding the two stacks together.

[0021] With continuing reference to FIG. **1** and with further reference to FIG. **2**, in some illustrative embodiments the color filter layer **12** is made up of photonic crystals **20**R, **20**G, **20**B where each photonic crystal comprises a two-dimensional (2D) array **40** of features **42** formed in a dielectric layer **44**, as diagrammatically shown in FIG. **2**. The dielectric layer **44** has a first refractive index, denoted herein as n_1, which may vary as a function of wavelength due to dispersion but is usually relatively constant over the visible spectrum of about 400-700 nm. For example, the refractive index of SiO_2 varies from about 1.47 at 400 nm to about 1.45 at 700 nm. The dielectric layer is typically an oxide layer such as a silicon oxide layer, for example stoichiometric SiO_2 or a silicon oxide of another stoichiometry. More generally, the dielectric layer **44** can be any dielectric material which is suitably translucent or transparent over the visible spectrum, such as silicon nitride (e.g. stoichiometric Si_3N_4), hafnium oxide (e.g. stoichiometric HfO_2), or so forth. The features **42** of each 2D array **40** of features **42** may, for example, comprise air or a filler dielectric material (that is different from the dielectric material of the dielectric layer **44**). In another embodiments, the features **42** may comprise the electroluminescent material **10**, which can be convenient from a manufacturing standpoint as the features comprising electroluminescent material **10** can be formed by filling openings etched into the dielectric layer **44** with the electroluminescent material **10** in a single spin coating or other deposition step that also deposits the electroluminescent material **10** as a layer on the color filter layer **12** as shown in FIG. **1**. The features **42** of each 2D array **40** have a second refractive index, denoted herein as n_2, which is different from the first refractive index n_1 of the dielectric layer **44**. Put another way, $n_1 \neq n_2$. Again, the second refractive index may vary as a function of wavelength due to dispersion but is usually relatively constant over the visible spectrum of about 400-700 nm. For example, the refractive index of air is about 1.00 over the visible spectrum.

[0022] With reference to FIGS. **3**, **4**, and **5**, an illustrative example of a generic photonic crystal **20** comprising a 2D array **40** of features **42** formed in a dielectric layer **44** is shown. The illustrative 2D array **40** of FIG. **3** is a hexagonal array **40** of features **42**; however, the array could have other geometries (for example, rectangular arrays as shown for the photonic crystals **20**R, **20**G, and **20**B of FIG. **2**). The features **42** of each 2D array **40** of features **42** forming a photonic crystal are periodically spaced, so that the photonic crystal is tuned to a resonant wavelength by a periodicity of the 2D array **40** of features **42**. The 2D periodicity of the array **40** of features **42** is two-dimensional in that the features **42** have inter-feature spacing that is periodic in two non-parallel in-plane directions, such as in x- and y-directions. This is shown in the enlarged view of FIG. **4**, as a spacing L, along an x-direction and a spacing Ly along a y-direction that is orthogonal to the x-direction in the plane of the dielectric layer **44**. In some embodiments, the periodicities may be written as:

$$L_x = \frac{m_x}{2}\frac{\lambda_{res}}{n_{pc}} \text{ where } m_x = 1, 3, 5, \ldots \tag{1}$$

and

$$L_y = \frac{m_y}{2}\frac{\lambda_{res}}{n_{pc}} \text{ where } m_y = 1, 3, 5, \ldots \tag{2}$$

[0023] where Ares is the resonant wavelength in a vacuum (which is equivalent to the resonant wavelength in air since the refractive index n_{air} of air is suitably close to the ideal refractive index n=1 for a vacuum), n_{pc} is the refractive index of the photonic crystal (which is typically about equal to the first refractive index n_1 of the dielectric material, neglecting the effect of the second refractive index n_2 of the features of the array of features on the photonic crystal medium), and m_x and m_y are odd positive integers. If the dispersion of n_{pc} (that is, its value as a function of wavelength) cannot be neglected, then in designing a photonic crystal for each of the red, green, and blue (or each of other primary colors) the value of n_{pc} for that wavelength is suitably used. In other words, the value n_{pc} ($\lambda_{res,R}$) is suitably used when selecting the periodicities Lx and Ly for the red photonic crystal **20**R (where $\lambda_{res,R}$ is the resonant wavelength of red light), and analogously for the green and blue photonic crystals **20**G and **20**B. In some embodiments, m_x=m_y, however this may not be the case in some other embodiments.

[0024] As previously discussed with reference to Equations (1) and (2), the periodicity of the 2D array **40** of features **42** beneficially provides for the photonic crystal (for example, generic photonic crystal **20** of FIG. **3**) to be tuned to the target resonant wavelength λ_{res} by the periodicity of the 2D array **40** of features **42**. However, in the illustrative embodiments, this periodicity is removed in a central region of the 2D array **40** of features **42**, thus forming a cavity within which the light (e.g. light **52**R, **52**G, or **52**B of FIG. **2**) is strongly confined.

[0025] With reference to FIG. **5**, this is illustrated for the hexagonal photonic crystal **20** of FIG. **3**. FIG. **5** illustrates the same enlarged perspective view of the central portion of the photonic crystal of FIG. **3** that is shown in FIG. **4**, but with labeling of the dielectric layer **44** and the features **42**. As labeled in FIG. **5**, a central cavity **46** is delineated by a boundary **47**, and a peripheral zone or region **48** is delineated as the portion of the photonic crystal **20** lying outside of the boundary **47**. In the core of the central cavity **46**, the features **42** are omitted, thus forming a cavity of the photonic crystal **20** within which the light (e.g. light **52**R, **52**G, or **52**B of FIG. **2**) is strongly confined by the features **42**. The light output by the electroluminescent material **10** is enhanced significantly by resonant coupling within the central cavity **46**, thus substantially increasing the intensity of the light at the resonant wavelength Ares of the photonic crystal **20** and providing the desired spectral filtering to Ares.

[0026] In the illustrative embodiment, the boundary **47** is hexagonal, however depending on the photonic crystal design this boundary could have a circular, square, or other geometry. The central cavity **46** may in some embodiments includes relatively few of the features **42**, optionally with a central-most portion containing none of the features as shown, and the features **42** that are in the central cavity **46** may be staggered as shown. The outer zone or region **48** contains the features **42** with the specified periodicity, e.g. as designed using Equations (1) and (2). It is to be appreciated that this is merely an illustrative example.

[0027] With returning reference to FIGS. **1** and **2**, each photonic crystal **20**R, **20**G, and **20**B (or, more generically, for the photonic crystal **20** of FIGS. **3-5**) comprises a corresponding 2D array **40** of features **42** formed in the dielectric layer **44**, which is tuned to the target resonant wavelength λ_{res} (e.g., λ_{res} is a red wavelength for a red photonic crystal **20**R, λ_{res} is a green wavelength for a green photonic crystal **20**G, and λ_{res} is a blue wavelength for a blue photonic crystal **20**B) by the periodicity Ly and Ly of the two-dimensional array **40** of features **42** making up that photonic crystal. Equations (1) and (2) can be used to design the periodicity for a given resonant wavelength λ_{res} and the refractive index n_{pc} of the photonic crystal (and more particularly n_{pc} at the specific target λ_{res} if the refractive index dispersion is sufficient to call for using a wavelength-specific refractive index). When the electrodes **22**, **24** electrically energize the electroluminescent material **10** to emit white light **50** (see FIG. **2**) that is optically coupled with a photonic crystal, the white light **50** is laterally confined in the plane of the dielectric layer **44** by the 2D array **40** of features **42**—that is, the 2D array **40** of features **42** forms a photonic crystal. There is no analogous confinement in the direction transverse to the plane of the dielectric layer **44** since there is no periodicity in that direction. (Some limited optical confinement could be present due to any abrupt refractive index step at the surfaces of the dielectric layer **44**; however, if this is a problem then anti-reflection coatings could be added to these surfaces to further reduce or eliminate optical confinement transverse to the plane of the dielectric layer **44**).

[0028] With continuing reference to FIG. **2**, confined resonant light in the plane of the dielectric layer **44** within the photonic crystals is diagrammatically depicted by arrows. Particularly, the red photonic crystal **20**R is resonant for red light **52**R. The green photonic crystal **20**G is resonant for green light **52**G. The blue photonic crystal **20**B is resonant for blue light **52**B. As already described with reference to Equations (1) and (2), the wavelength λ_{res} of the light which is resonant for each photonic crystal depends on the periodicity of the features **42** of the array **40** of features **42** making up that photonic crystal. The strength of the optical resonance provided by the photonic crystal depends in part on the difference between the first refractive index n_1 of the host dielectric layer **44** and the second refractive index n_2 of the features **42**. This difference can be quantified, for example, using the refractive index contrast A given by:

$$\Delta = \frac{(n_1^2 - n_2^2)}{2n_1^2} \tag{3}$$

[0029] A larger value for the refractive index contrast A generally leads to stronger resonance and improved color selection by the photonic crystal. The photonic crystal can suitably operate with either $n_1 > n_2$ or with $n_1 < n_2$.

[0030] With reference to FIGS. **6** and **7**, moreover, the first-color photonic crystals (e.g. red photonic crystals **20**R), the second-color photonic crystals (e.g. green photonic crystals **20**G), and the third-color photonic crystals (e.g. blue photonic crystals **20**B) are suitably arranged to form a pixel **60** (FIG. **6**) or pixel **62** (FIG. **7**) that includes at least one first-color photonic crystal **20**R, at least one second-color photonic crystal **20**G, and at least one third-color photonic crystal **20**B. The pixel **60** of FIG. **6** has photonic crystals formed by hexagonal arrays **40** of features **42** (such as that shown in FIG. **3**), while the pixel **62** of FIG. **7** has photonic crystals formed by rectangular arrays **40** of features **42** (such as that shown in FIG. **2**). These are merely illustrative examples. Moreover, while the illustrative pixels **60** and **62** have one red photonic crystal **20**R, one green photonic crystal **20**G, and one blue photonic crystal **20**B, it is contemplated to include more than one of selected crystals—for example, some display designs may have each pixel include two green photonic crystals **20**G to improve the perceived color balance.

[0031] Furthermore, as indicated by dashed lines in FIGS. **6** and **7**, it is contemplated to have each pixel **60** or **62** further include a white element **64** to provide a higher quality of white light than might be achieved by operating the red, green, and blue "sub-pixels" together. The white element **64**, if included, can be readily implemented by not including any array of features in the area of the dielectric layer **44** corresponding to the white element **64**. In this way, the white light **50** is not modified by any photonic crystal in that area and hence is output as high quality white light. (In such a design, the TFT array backplane **30** in an active matrix design suitably include an additional TFT for controlling emission of the white light **50** at the location of the white element **64**). As another contemplated variant, instead of adding the white element **64** the pixel could incorporate an additional color element, i.e. a fourth-color photonic crystal tuned to a fourth resonant wavelength different from the first, second, and third resonant wavelengths, or even a fifth-color photonic crystal or more.

[0032] With reference to FIG. **8**, it will be further understood that to implement the color display the first-color photonic crystals **20**R, the second-color photonic crystals **20**G, and the third-color photonic crystals **20**B are arranged to form an array of pixels **60** or **62** spanning a display area **66** of the color display. Illustrative example of FIG. **8** shows by way of nonlimiting illustrative example pixels **60** of FIG. **6** (without the optional white element **64**) spanning the display area **66**; however, the pixels could be otherwise designed, such as the rectangular pixels **62** of FIG. **7**, and/or the pixels can include the white element **64**. FIG. **8** is diagrammatic, and it will be appreciated that in typical color displays for televisions, color computer displays, or the like the number of pixels **60** is much larger than shown, e.g. a typical computer display may have a 1920×1080 array of pixels **60**, as one specific non-limiting illustrative example.

[0033] With reference back to FIGS. **3**-**5** and with further reference to FIGS. **9**, **10**, and **11**, some suitable approaches for fabricating the color filter layer **12** including the features **42** are described. FIGS. **9**-**11** depict Section S-S indicated in FIG. **4** and corresponding to a plane oriented perpendicular to the plane of the dielectric layer **44** (and hence also perpendicular to the plane of the color filter layer **12**) and that passes through three features **42**, according to three different embodiments.

[0034] With particular reference to FIG. **9**, in this embodiment the features **42** are cylindrical features **42** comprising cylindrical openings **70** passing through the dielectric layer **44**, with a central axis **72** of each opening oriented perpendicular to the plane **74** of the dielectric layer **44**. The cylindrical openings **70** can have any cross-sectional shape, e.g. circular, oval, square, rectangular, hexagonal, or so forth. The cylindrical openings **70** can, for example, be formed by photolithographically controlled etching of the dielectric layer **44**. Also shown in FIG. **9** is the electroluminescent material **10** disposed as a coating on the color filter layer **12**. In the embodiment of FIG. **9**, the features **42** comprise the cylindrical openings **70** filled with air. This can be achieved if, for example, the electroluminescent material **10** is sufficiently viscous and the cross-sectional diameter of the cylindrical openings **70** is small enough so that the electroluminescent material **10** does not flow into the openings **70**. In another approach, the air-filled openings **70** can be formed after deposition of the electroluminescent material **10**. Because the openings **70** are filled with air, in this embodiment the second refractive index for the features **42** is the refractive index of air, that is, $n_2=1.00$.

[0035] With particular reference to FIG. **10**, in this embodiment the features **42** comprise filler dielectric material **80** forming cylindrical features **42** passing through the dielectric layer **44**. The cylindrical filler dielectric material features **42** of this embodiment have cylinder axes **72** again oriented perpendicular to the plane **74** of the dielectric layer. These cylindrical filler dielectric material features **42** may be formed by photolithographically controlled etching of the dielectric layer **44** to form cylindrical openings analogous to the cylindrical openings **70** of the embodiment of FIG. **9**, followed by a deposition process in which the openings are filled with the filler dielectric material **80**. In one suitable approach, this entails a spin coating, sputtering, or other suitable deposition of the filler dielectric material **80** followed by chemical-mechanical polishing (CMP) to remove any filler dielectric material coating the surface of the dielectric layer **44**. An advantage of this embodiment is that the second refractive index n_2 is the refractive index of the filler dielectric material **80**, and hence the second refractive index n_2 is amenable to design by choice of the filler dielectric material **80**. For example, if the dielectric layer **44** is a low-k dielectric material such as silicon dioxide (SiO_2, for which $n_1=1.47$ at 400 nm to about $n_1=1.45$ at 700 nm) then the filler dielectric material **80** could be a high-k dielectric material such as silicon nitride (Si_3N_4, yielding $n_2 \simeq 2.0$-2.1 depending on wavelength).

[0036] With particular reference to FIG. **11**, in this embodiment the features **42** comprise portions of the electroluminescent material **10** filling openings that pass through the dielectric layer **44** (analogous to the openings **70** of the embodiment of FIG. **9**). The features **42** comprising cylindrical portions of the translucent material **10** of this embodiment have cylinder axes **72** yet again oriented perpendicular to the plane **74** of the dielectric layer. These cylindrical portions of the translucent material forming the features **42** may be formed by photolithographically controlled etching of the dielectric layer **44** to form cylindrical openings analogous to the cylindrical openings **70** of the embodiment of FIG. **9**, followed by deposition of the electroluminescent material **10** over the surface of the dielectric layer **44** such a way that the deposited electroluminescent material **10** also flows into and fills those cylindrical openings. To achieve such filling, the electroluminescent material **10** is sufficiently fluid and the cross-sectional diameter of the cylindrical openings is large enough so that the electroluminescent material **10** flows into the openings. In this embodiment, the second refractive index n_2 is the refractive index of the electroluminescent material **10**. An advantage of this embodiment is that no additional step is performed to fill the openings (versus the embodiment of FIG. **10** which entails the additional deposition/CMP to deposit the filler dielectric material **80** in the openings). Another advantage of this embodiment is that the portions of the electroluminescent material **10** disposed in the openings passing through the dielectric layer **44** can be energized by the electrodes **22**, **24** (see FIG. **1**) so as to increase to total volume of the electroluminescent material **10** providing the light **50** (see FIG. **2**).

[0037] With reference to FIG. **12**, a flow chart is shown of a manufacturing method for forming a multicolor light emission structure of a color display employing an electroluminescent material emitting white light and including a color filter layer made up of photonic crystals comprising 2D arrays of features formed in a dielectric layer. For example the manufacturing method of FIG. **12** could be used to manufacture the light emission structure of FIG. **1** including the electroluminescent layer **10** and the color filter layer **12** as previously described. The method starts with forming a dielectric layer (for example, the dielectric layer **44**). This may be done, for example, by depositing silicon dioxide (SiO_2) or another chosen dielectric material on a suitable substrate such as a glass or sapphire substrate or the like. In some embodiments the substrate may have a previously formed TFT backplate or other electronics layer constituting the control circuitry **30** diagrammatically shown in FIG. **1**.

[0038] In some embodiments, the operation **100** deposits the dielectric layer **44** with uniform thickness. As a consequence, the resulting color filter layer **12** will also be of uniform thickness. This provides a planar surface on which subsequent layers of the color display are deposited, which can simplify manufacturing and improve reliability and/or performance of the manufactured color display.

[0039] In an operation **102**, a two-dimensional array of openings is formed in the dielectric layer. These openings may constitute the features **42** of the 2D array **40** of features **42** as in the embodiment of FIG. **9**. Alternatively, in an optional operation **104**, after forming the openings they may be filled with filler dielectric material, e.g. with filler dielectric material **80** forming cylindrical features **42** passing through the dielectric layer **44** as in the embodiment of FIG. **10**.

[0040] In an operation **106**, electroluminescent material is formed on the dielectric layer. For example, the electroluminescent material formed in the operation **106** may be the electroluminescent material **10** as previously described. The operation **106** may deposit the electroluminescent material **10** on the dielectric layer using spin coating or another suitable approach. In a nonlimiting illustrative embodiment, the material deposited by spin coating or the like to form the electroluminescent material **10** as an OLED material comprises a liquid polymer containing fluorescent or phosphorescent dye, polymer, or other suitable molecule. In another nonlimiting illustrative embodiment, the material deposited by spin coating or the like to form the electroluminescent material **10** comprises a liquid polymer containing colloidal quantum dots (QDs) so as to form the electroluminescent material as a colloidal QD electroluminescent material. Optionally, the operation **106** may form the electroluminescent material as a multilayer stack, for example including electron and/or hole injection or transport layer(s). For embodiments in accordance with FIG. **11**, the operation **106** also results in the electroluminescent material filling the openings made in operation **102**, as indicated by block **108** of FIG. **12**, so as to form the features **42** of the 2D array **40** of features **42** as cylindrical columns of the electroluminescent material **10** passing through the dielectric layer.

[0041] With continuing reference to FIG. **12**, an advantage of the above-described manufacturing process is that it can be performed in some embodiments using only a single photolithography mask **110** for the operation **102** to form all of the 2D arrays **40** of features **42** of all photonic crystals of the color filter layer **12**. By contrast, if the color filter layer is constructed using, for example, dielectric layers of different thicknesses for the different resonant wavelengths (e.g. red, green, and blue), then multiple photomasks will typically be used to form the regions with these different thicknesses. Moreover, the approach of FIG. **12** is readily scalable to form multicolor light emission structures for color displays of different display resolutions, by employing an embodiment of the photomask **110** for the desired display resolution. Still further, as the wavelength selection provided by the photonic crystals is determined by the periodicity of the 2D arrays of features which is in turn determined by the photomask **110**, this results in the manufacturing process of FIG. **12** providing precise run-to-run reproducibility in defining the emission wavelengths.

[0042] With returning reference to FIG. **1**, the embodiments described thus far employ the white electroluminescent material **10** which outputs white light when energized by the electrodes **22**, **24**. In these designs, the red, green, and blue (or other primary colors) of the color display are achieved by wavelength selection provided by the photonic crystals **20**R, **20**G, and **20**B.

[0043] With reference to FIG. **13**, another color display embodiment is similar to that of FIG. **1**, except that in the embodiment of FIG. **13** the white electroluminescent material **10** is replaced by regions of red electroluminescent material **10**R, regions of green electroluminescent material **10**G, and regions of blue electroluminescent material **10**B (or, more generally, by regions of first-color electroluminescent material **10**R, regions of second-color electroluminescent material **10**G, and regions of third-color electroluminescent material **10**B). As shown in FIG. **13**, the regions of red electroluminescent material **10**R are aligned with corresponding red photonic crystals **20**R, the regions of green electroluminescent material **10**G are aligned with corresponding green photonic crystals **20**G, and the regions of blue electroluminescent material **10**B are aligned with corresponding blue photonic crystals **20**B. Operation of this embodiment is analogous to that of FIG. **1**, except that in this case the photonic crystals serve to improve the color purity of the red, green, and blue light output by the respective red, green, and blue electroluminescent material **10**R, **10**G, and **10**B.

[0044] In the following, some further embodiments are described.

[0045] In a nonlimiting illustrative embodiment, a color display comprises: a color filter layer comprising an array of photonic crystals formed in a dielectric layer having a first refractive index; an electroluminescent material optically coupled with the color filter layer; and electrodes arranged to electrically energize the electroluminescent material to output light. Each photonic crystal includes a 2D array of features. The features have a second refractive index different from the first refractive index. The photonic crystal is tuned to a resonant wavelength by a periodicity of the 2D array of features. The array of photonic crystals includes first-color photonic crystals tuned to a first resonant wavelength, second-color photonic crystals tuned to a second resonant wavelength that is different from the first resonant wavelength, and third-color photonic crystals tuned to a third resonant wavelength that is different from the first resonant wavelength and that is different from the second resonant wavelength. The first-color photonic crystals, the second-color photonic crystals, and the third-color photonic crystals are arranged to form an array of pixels spanning a display area of the color display, in which each pixel includes at least one first-color photonic crystal, at least one second-color photonic crystal, and at least one third-color photonic crystal.

[0046] In a nonlimiting illustrative embodiment, a method of fabricating a color display comprises forming a dielectric layer over a display area of the color display wherein the dielectric layer has a first refractive index, forming 2D arrays of features in the dielectric layer, and coating the dielectric layer with an electroluminescent material. The 2D arrays of features formed in the dielectric layer include: first 2D arrays of features having a first periodicity and forming first-color photonic crystals tuned to a first resonant wavelength; second 2D arrays of openings having a second periodicity that is different from the first periodicity and forming second-color photonic crystals tuned to a second resonant wavelength different from the first resonant wavelength; and third 2D arrays of openings having a third periodicity that is different from the first periodicity and that is different from the second periodicity and forming third-color photonic crystals tuned to a third resonant wavelength different from the first resonant wavelength and different from the second resonant wavelength.

[0047] In a nonlimiting illustrative embodiment, a color display comprises a color filter layer including a dielectric layer with a first refractive index and with an array of photonic crystals formed therein, an electroluminescent material disposed on the color filter layer, and electrodes arranged to electrically energize the electroluminescent material to cause the electroluminescent material to output white light. Each photonic crystal includes a 2D array of features. The features have a second refractive index different from the first refractive index. The 2D array of features includes a central cavity within which the features of the 2D array of features are omitted. Each photonic crystal is tuned to a resonant wavelength by a periodicity of the two-dimensional array of features. The array of photonic crystals include red photonic crystals tuned to a red resonant wavelength, green photonic crystals tuned to a green resonant wavelength, and blue photonic crystals tuned to a blue resonant wavelength. The red photonic crystals, the green photonic crystals, and the blue photonic crystals are arranged to form an array of pixels spanning a display area of the color display, in which each pixel includes at least one red photonic crystal, at least one green photonic crystal, and at least one blue photonic crystal.

[0048] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A color display, comprising:

a substrate; and

an optical stack on the substrate, the optical stack comprising:

a color filter layer including a dielectric layer having a first refractive index, and a photonic crystal formed in the dielectric layer and tuned to a resonant wavelength, wherein the photonic crystal includes a two-dimensional array of features having a second refractive index different from the first refractive index;

an electroluminescent layer on the substrate that is optically coupled with the photonic crystal; and

electrodes comprising an anode and a cathode arranged to electrically energize the electroluminescent layer to output light;

wherein the color filter layer is disposed between the anode and the cathode.

2. The color display of claim **1**, wherein the color display is a bottom-emitting display or a top-emitting display.

3. The color display of claim **1**, further comprising a backplane comprising control circuitry disposed between the substrate and the optical stack.

4. The color display of claim **1**, wherein the substrate comprises glass or sapphire.

5. The color display of claim **1**, wherein the photonic crystal comprises an array of photonic crystals.

6. The color display of claim **5**, wherein the array of photonic crystals includes first-color photonic crystals tuned to a first resonant wavelength, second-color photonic crystals tuned to a second resonant wavelength that is different from the first resonant wavelength, and third-color photonic crystals tuned to a third resonant wavelength that is different from the first resonant wavelength and that is different from the second resonant wavelength.

7. The color display of claim **6**, wherein the first-color photonic crystals, the second-color photonic crystals, and the third-color photonic crystals are arranged to form an array of pixels spanning a display area of the color display, in which each pixel includes at least one first-color photonic crystal, at least one second-color photonic crystal, and at least one third-color photonic crystal.

8. The light emission structure of claim **7** wherein the electroluminescent layer emits white light when electrically energized by the electrodes.

9. The color display of claim **7**, wherein each pixel further includes a white element.

10. The color display of claim **9**, wherein the electroluminescent layer emits white light when electrically energized by the electrodes, and each white element comprises a region of the dielectric layer that does not include the features having the second refractive index.

11. The color display of claim **1**, wherein the electroluminescent layer comprises an organic electroluminescent diode (OLED) material or a colloidal quantum dot (QD) electroluminescent material.

12. The color display of claim **1** wherein the features having the second refractive index comprise cylindrical features that pass through the dielectric layer and have cylinder axes oriented perpendicular to a plane of the dielectric layer.

13. A method of operating a color display, comprising:

applying an electrical bias to selected pixels on the color display to form a desired color image;

wherein the color display comprises a color filter layer that comprises an array of photonic crystals formed in a dielectric layer having a first refractive index, an electroluminescent material optically coupled with the color filter layer, and electrodes arranged to electrically energize the electroluminescent material to output light;

wherein each photonic crystal includes a two-dimensional array of features, the features having a second refractive index different from the first refractive index, the photonic crystal being tuned to a resonant wavelength by a periodicity of the two-dimensional array of features;

wherein the array of photonic crystals includes first-color photonic crystals tuned to a first resonant wavelength, second-color photonic crystals tuned to a second resonant wavelength that is different from the first resonant wavelength, and third-color photonic crystals tuned to a third resonant wavelength that is different from the first resonant wavelength and that is different from the second resonant wavelength; and

wherein the first-color photonic crystals, the second-color photonic crystals, and the third-color photonic crystals are arranged to form an array of pixels spanning a display area of the color display, in which each pixel includes at least one first-color photonic crystal, at least one second-color photonic crystal, and at least one third-color photonic crystal.

14. The method of claim **13**, wherein the electrical bias is applied using a thin-film transistor (TFT) array, and the electrodes are segmented to form addressable pixels of the color display each of which is independently electrically biased or not biased.

15. The method of claim **14**, wherein the TFTs of the TFT array are polycrystalline silicon (poly-Si) or amorphous silicon (a-Si) TFTs.

16. The method of claim **14**, wherein the color display includes a backplane comprising the TFT array and a substrate on which the backplane is disposed.

17. A method of operating a color display, comprising:

applying an electrical bias to the color display to form a desired image;

wherein the color display comprises:

a dielectric layer having a first refractive index;

a photonic crystal formed in the dielectric layer and tuned to a resonant wavelength, wherein the photonic crystal includes a two-dimensional array of features having a second refractive index different from the first refractive index;

an electroluminescent layer optically coupled with the photonic crystal; and

electrodes comprising an anode and a cathode arranged to electrically energize the electroluminescent layer to output light;

wherein the dielectric layer and the photonic crystal formed in the dielectric layer are disposed between the anode and the cathode.

18. The method of claim **17**, wherein the electrical bias is applied using a thin-film transistor (TFT) array, and the electrodes are segmented to form addressable pixels of the color display each of which is independently electrically biased or not biased.

19. The method of claim **18**, wherein the TFTs of the TFT array are polycrystalline silicon (poly-Si) or amorphous silicon (a-Si) TFTs.

20. The method of claim **18**, wherein the color display includes a backplane comprising the TFT array and a substrate on which the backplane is disposed.

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